Course Project

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Course title: Agile Development Process and Devops

Slot: C2+TC2

Course code: ISWE406P

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Topic: Task Manager

Create basic python project structure

```python
from flask import Flask, request, jsonify

app = Flask(__name__)
tasks = []

@app.route('/add', methods=['POST'])
def add_task():
    task = request.json.get('task')
    tasks.append({"task": task, "status": "pending"})
    return jsonify(tasks)

@app.route('/')
def home():
    return "Task Manager v1 Running"

if __name__ == '__main__':
    app.run(host='0.0.0.0', port=5000)
```

Codes:

Version 1:

Index.html:

```html
<!DOCTYPE html>
<html>
<head>
    <title>Task Manager v1</title>
</head>
<body>

<h2>Task Manager v1 (Add Only)</h2>

<form method="POST" action="/add">
    <input type="text" name="task" placeholder="Enter task">
    <button type="submit">Add</button>
</form>

<ul>
{% for task in tasks %}
    <li>{{ task }}</li>
{% endfor %}
</ul>

</body>
</html>
```

App.py

```python
@app.route('/')
def index():
    return render_template_string(HTML, version=APP_VERSION)

@app.route('/tasks', methods=['GET'])
def get_tasks():
    return jsonify(tasks)

@app.route('/tasks', methods=['POST'])
def create_task():
    data = request.json
    task = {"id": len(tasks)+1, "title": data["title"], "description": data.get("desc
    tasks.append(task)
    return jsonify(task), 201

if __name__ == '__main__':
    app.run(host='0.0.0.0', port=5000)
```

Dockerfile:

```
FROM python:3.9
WORKDIR /app
COPY . .
RUN pip install -r requirements.txt
CMD ["python", "app.py"]
```

Requirements.txt

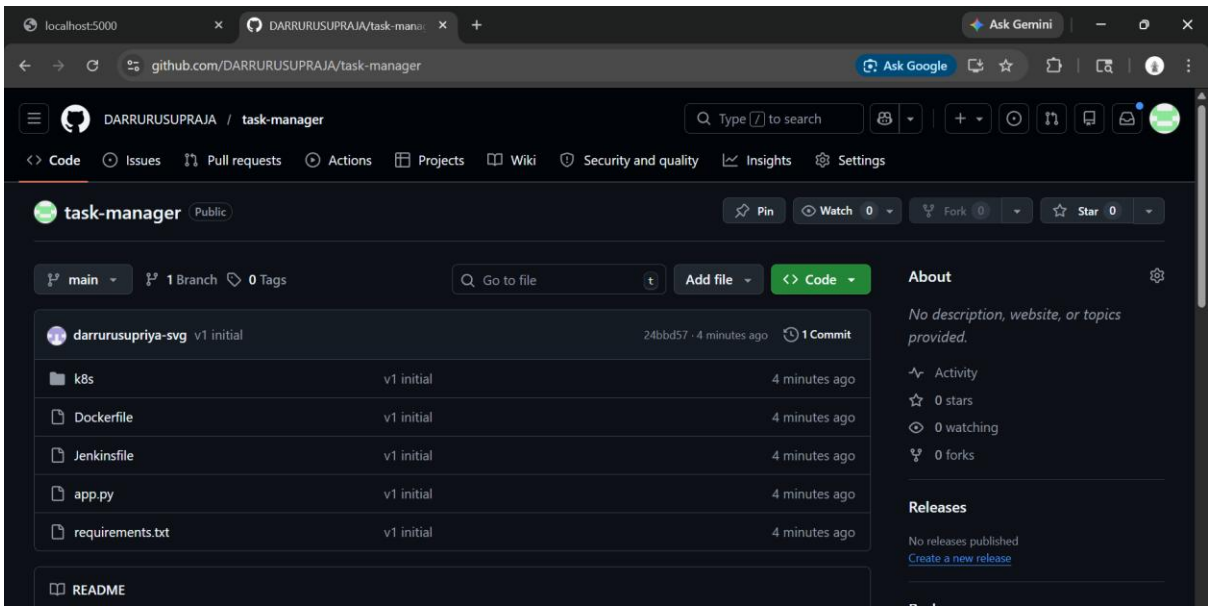
```
flask
```

V1 deployment.yaml:

v1-deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: task-manager-v1
spec:
  replicas: 1
  selector:
    matchLabels:
      app: task-manager-v1
  template:
    metadata:
      labels:
        app: task-manager-v1
    spec:
      containers:
      - name: task-manager
        image: darrurusupraja/task-manager:v1
        ports:
        - containerPort: 5000
---
apiVersion: v1
kind: Service
metadata:
  name: task-manager-v1-service
spec:
  type: NodePort
  selector:
    app: task-manager-v1
  ports:
    - port: 5000
      targetPort: 5000
      nodePort: 30001
```

```
PS C:\Users\HP\OneDrive\Desktop\task-manager> git checkout -b v1
Switched to a new branch 'v1'
PS C:\Users\HP\OneDrive\Desktop\task-manager> git push origin v1
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'v1' on GitHub by visiting:
remote:      https://github.com/DARRURUSUPRAJA/task-manager/pull/new/v1
remote:
To https://github.com/DARRURUSUPRAJA/task-manager.git
 * [new branch]      v1 -> v1
PS C:\Users\HP\OneDrive\Desktop\task-manager>
```



Version 2:

Index.html:

```
v2 > templates > index.html > html
<!DOCTYPE html>
<html>
<head>
    <title>Task Manager v2</title>
</head>
<body>

<h2>Task Manager v2 (Add + Update)</h2>

<form method="POST" action="/add">
    <input type="text" name="task" placeholder="Enter task">
    <button type="submit">Add</button>
</form>

<ul>
{% for t in tasks %}
    <li>
        {{ t.task }} - {{ t.status }}
        <a href="/update/{{ loop.index0 }}">✓</a>
    </li>
{% endfor %}
</ul>
```

```
22    </ul>
23
24    </body>
25    </html>
```

App.py:

```python
@app.route('/')
def home():
    return render_template_string(HTML, version=APP_VERSION)

@app.route('/tasks', methods=['GET'])
def get_tasks():
    return jsonify(tasks)

@app.route('/tasks', methods=['POST'])
def add_task():
    data = request.json
    task = {
        "id": len(tasks) + 1,
        "title": data["title"],
        "description": data.get("description", ""),
        "status": "TODO"
    }
    tasks.append(task)
    return jsonify(task)

@app.route('/tasks/<int:task_id>', methods=['PUT'])
def update_task(task_id):
    data = request.json
```

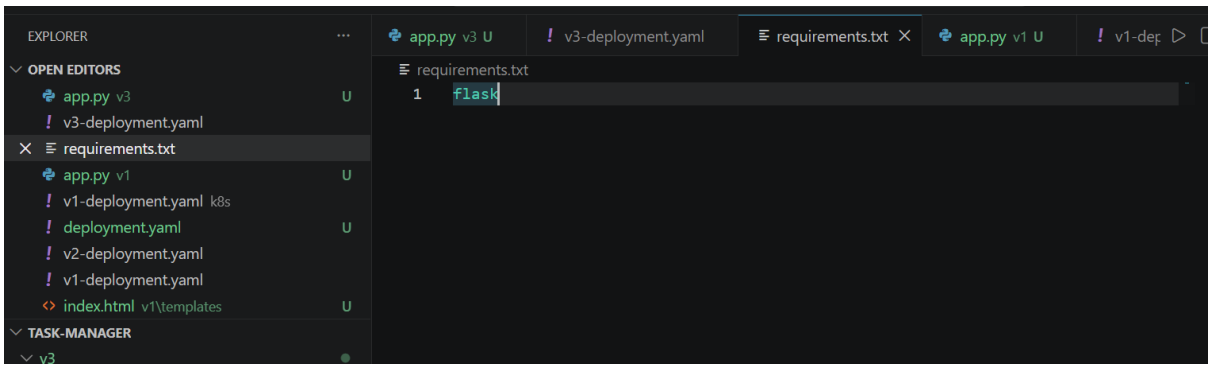
```python
    for task in tasks:
        if task["id"] == task_id:
            task["status"] = data["status"]
            return jsonify(task)
    return jsonify({"error": "Task not found"}), 404

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000, debug=True)
```

Dockerfile:

```dockerfile
FROM python:3.9
WORKDIR /app
COPY . .
RUN pip install -r requirements.txt
CMD ["python", "app.py"]
```

Requirements.txt

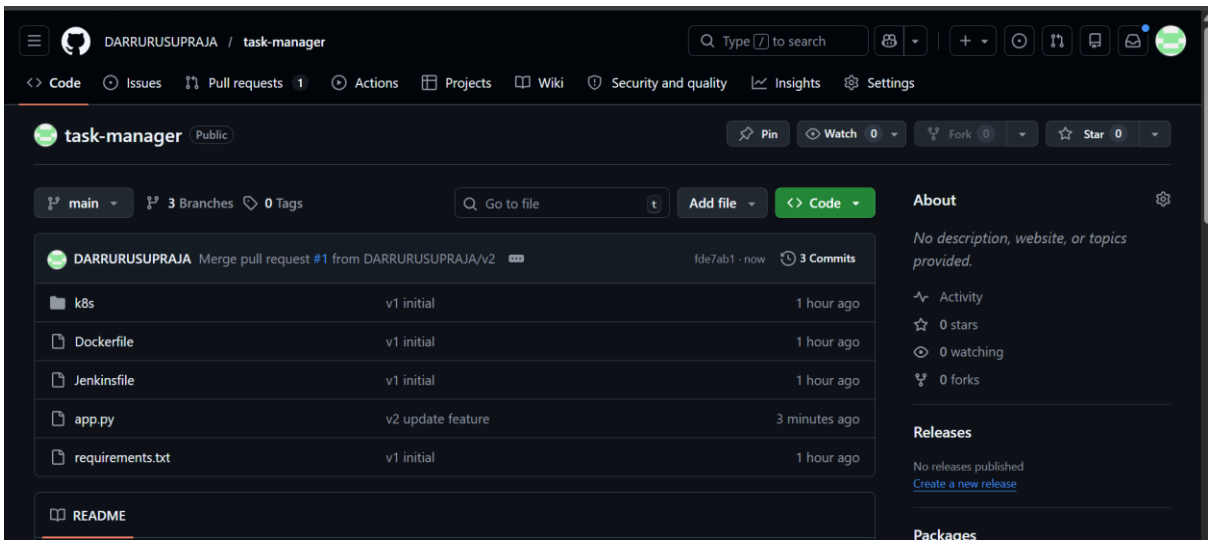


V2 deployment.yaml:

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: task-manager-v2
spec:
  replicas: 1
  selector:
    matchLabels:
      app: task-manager-v2
  template:
    metadata:
      labels:
        app: task-manager-v2
    spec:
      containers:
      - name: task-manager
        image: darrurusupraja/task-manager:v2
        ports:
        - containerPort: 5000
---
apiVersion: v1
kind: Service
metadata:
```

```yaml
  name: task-manager-v2-service
spec:
  type: NodePort
  selector:
    app: task-manager-v2
  ports:
    - port: 5000
      targetPort: 5000
      nodePort: 30002
```

```
PS C:\Users\HP\OneDrive\Desktop\task-manager> git checkout -b v2
Switched to a new branch 'v2'
PS C:\Users\HP\OneDrive\Desktop\task-manager> git commit -am "v2 update feature"
[v2 dbf7984] v2 update feature
 1 file changed, 8 insertions(+), 1 deletion(-)
PS C:\Users\HP\OneDrive\Desktop\task-manager> git push origin v2
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 437 bytes | 18.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
remote:
remote: Create a pull request for 'v2' on GitHub by visiting:
remote:      https://github.com/DARRURUSUPRAJA/task-manager/pull/new/v2
remote:
To https://github.com/DARRURUSUPRAJA/task-manager.git
 * [new branch]      v2 -> v2
PS C:\Users\HP\OneDrive\Desktop\task-manager>
```



Version 3:

Index.html:

v3 > templates > index.html > html

```html
<!DOCTYPE html>
<html>
<head>
    <title>Task Manager v3</title>
</head>
<body>

<h2>Task Manager v3 (Full)</h2>

<form method="POST" action="/add">
    <input type="text" name="task" placeholder="Enter task">
    <button type="submit">Add</button>
</form>

<h3>Pending Tasks</h3>
<ul>
{% for task in tasks %}
    <li>
        {{ task }}
        <a href="/complete/{{ loop.index0 }}">✓</a>
        <a href="/delete/{{ loop.index0 }}">❌</a>
    </li>
{% endfor %}
</ul>

<h3>Completed Tasks</h3>
<ul>
{% for task in completed %}
    <li>{{ task }}</li>
{% endfor %}
</ul>

</body>
</html>
```

App.py:

```python
@app.route('/')
def index():
    return render_template_string(HTML, version=APP_VERSION)

@app.route('/tasks', methods=['GET'])
def get_tasks():
    return jsonify(tasks)

@app.route('/tasks', methods=['POST'])
def create_task():
    data = request.json
    if not data or not data.get('title'):
        return jsonify({"error": "Title is required"}), 400
    task = {
        "id": len(tasks) + 1,
        "title": data["title"],
        "description": data.get("description", ""),
        "status": "TODO"
    }
    tasks.append(task)
    return jsonify(task), 201

@app.route('/tasks/<int:task_id>', methods=['PUT'])
```

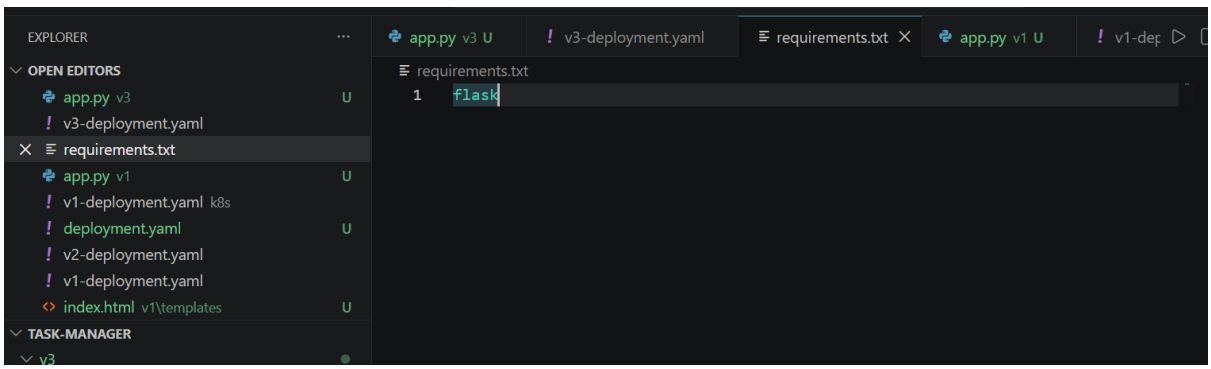
```python
def get_stats():
    return jsonify({
        "total": len(tasks),
        "TODO": len([t for t in tasks if t["status"] == "TODO"]),
        "IN_PROGRESS": len([t for t in tasks if t["status"] == "IN_PROGRESS"]),
        "DONE": len([t for t in tasks if t["status"] == "DONE"])
    })

if __name__ == '__main__':
    app.run(host='0.0.0.0', port=5000, debug=True)
```

Dockerfile:

```dockerfile
FROM python:3.9
WORKDIR /app
COPY . .
RUN pip install -r requirements.txt
CMD ["python", "app.py"]
```

Requirements.txt

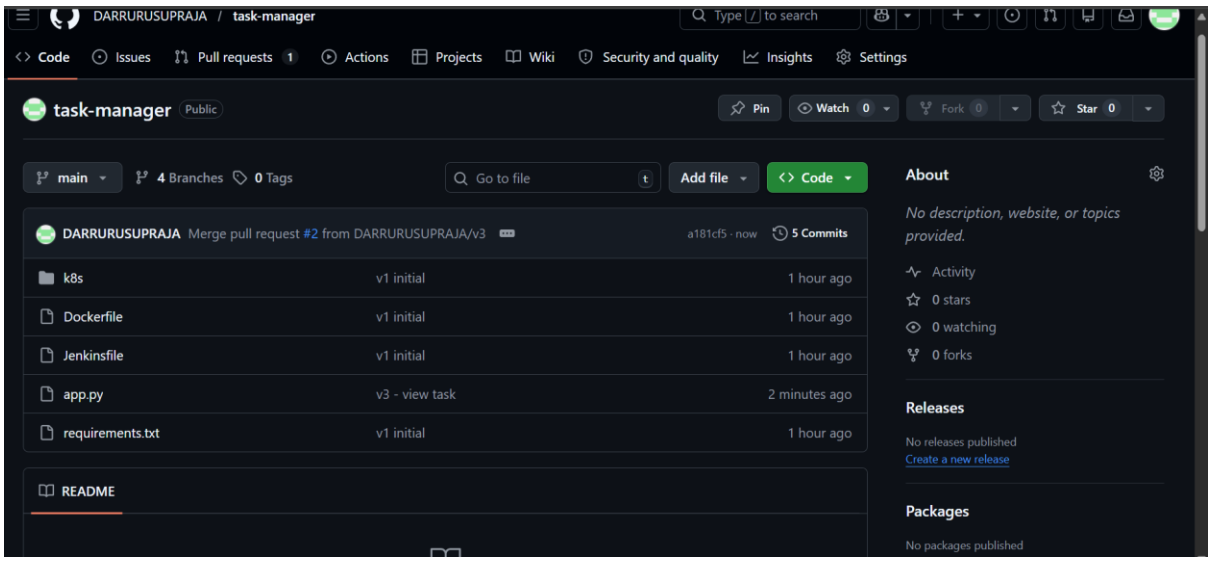


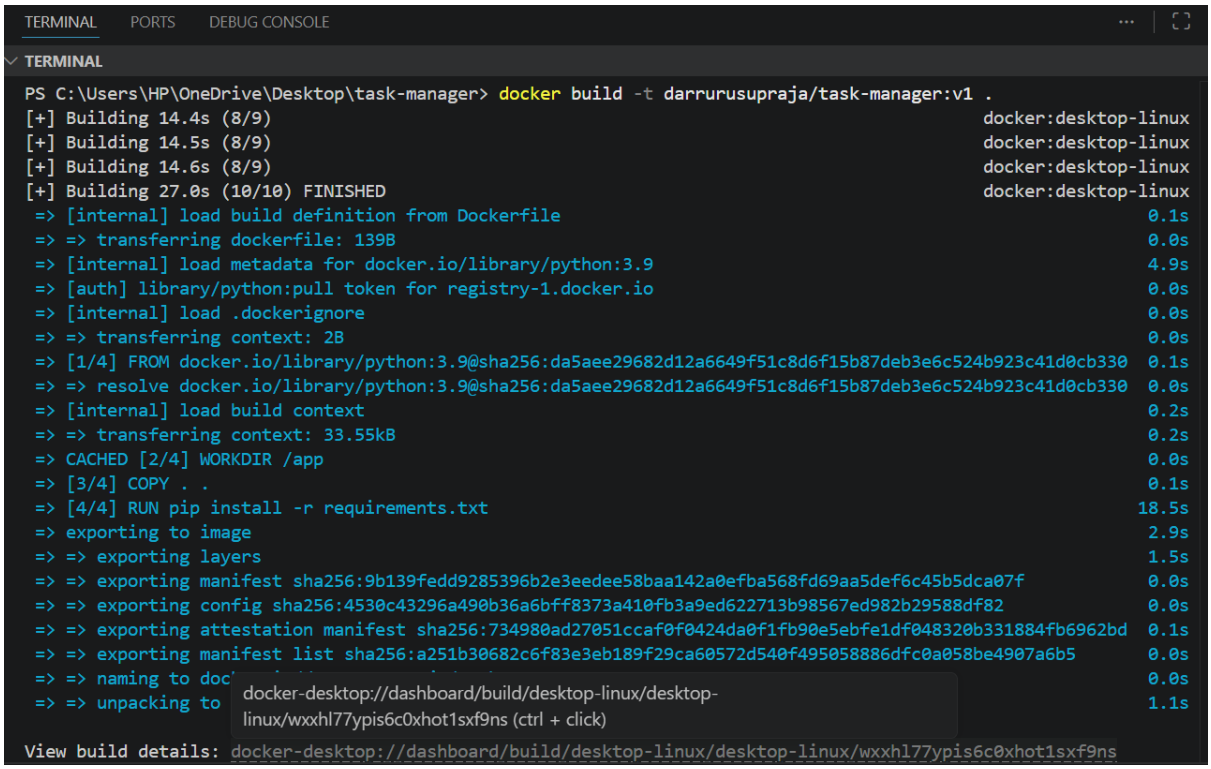
V3 deployment.yaml:

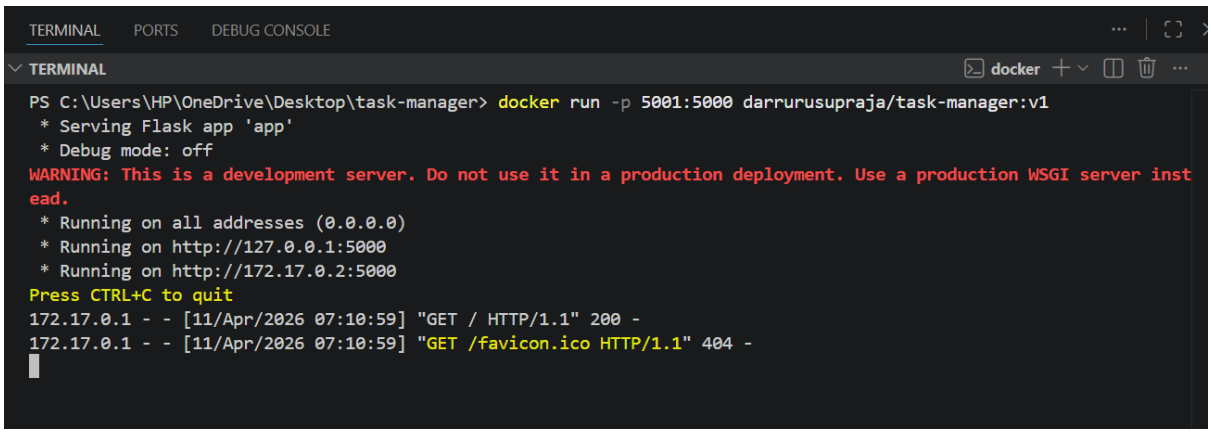
v3-deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: task-manager-v3
spec:
  replicas: 1
  selector:
    matchLabels:
      app: task-manager-v3
  template:
    metadata:
      labels:
        app: task-manager-v3
    spec:
      containers:
      - name: task-manager
        image: darrurusupraja/task-manager:v3
        ports:
        - containerPort: 5000
---
apiVersion: v1
kind: Service
metadata:
  name: task-manager-v3-service
spec:
  type: NodePort
  selector:
    app: task-manager-v3
  ports:
    - port: 5000
      targetPort: 5000
      nodePort: 30003
```

```
PS C:\Users\HP\OneDrive\Desktop\task-manager> git checkout -b v3
Switched to a new branch 'v3'
PS C:\Users\HP\OneDrive\Desktop\task-manager> git commit -am "v3 - view task"
[v3 7271b3a] v3 - view task
 1 file changed, 21 insertions(+), 5 deletions(-)
PS C:\Users\HP\OneDrive\Desktop\task-manager> git push origin v3
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 561 bytes | 280.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
remote:
remote: Create a pull request for 'v3' on GitHub by visiting:
remote:      https://github.com/DARRURUSUPRAJA/task-manager/pull/new/v3
remote:
To https://github.com/DARRURUSUPRAJA/task-manager.git
 * [new branch]      v3 -> v3
PS C:\Users\HP\OneDrive\Desktop\task-manager>
```


DARRURUSUPRAJA / task-manager
Code
Issues
Pull requests 1
Actions
Projects
Wiki
Security and quality
Insights
Settings
task-manager Public
Pin
Watch 0
Fork 0
Star 0
main
4 Branches
0 Tags
Go to file
Add file
Code
DARRURUSUPRAJA Merge pull request #2 from DARRURUSUPRAJA/v3
a181cf5 · now
5 Commits
k8s v1 initial 1 hour ago
Dockerfile v1 initial 1 hour ago
Jenkinsfile v1 initial 1 hour ago
app.py v3 - view task 2 minutes ago
requirements.txt v1 initial 1 hour ago
README
About
No description, website, or topics provided.
Activity
0 stars
0 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published


TERMINAL PORTS DEBUG CONSOLE
TERMINAL
PS C:\Users\HP\OneDrive\Desktop\task-manager> docker build -t darrurusupraja/task-manager:v1 .
[+] Building 14.4s (8/9) docker:desktop-linux
[+] Building 14.5s (8/9) docker:desktop-linux
[+] Building 14.6s (8/9) docker:desktop-linux
[+] Building 27.0s (10/10) FINISHED docker:desktop-linux
=> [internal] load build definition from Dockerfile 0.1s
=> => transferring dockerfile: 139B 0.0s
=> [internal] load metadata for docker.io/library/python:3.9 4.9s
=> [auth] library/python:pull token for registry-1.docker.io 0.0s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [1/4] FROM docker.io/library/python:3.9@sha256:da5aee29682d12a6649f51c8d6f15b87deb3e6c524b923c41d0cb330 0.1s
=> => resolve docker.io/library/python:3.9@sha256:da5aee29682d12a6649f51c8d6f15b87deb3e6c524b923c41d0cb330 0.0s
=> [internal] load build context 0.2s
=> => transferring context: 33.55kB 0.2s
=> CACHED [2/4] WORKDIR /app 0.0s
=> [3/4] COPY . . 0.1s
=> [4/4] RUN pip install -r requirements.txt 18.5s
=> exporting to image 2.9s
=> => exporting layers 1.5s
=> => exporting manifest sha256:9b139fedd9285396b2e3eedee58baa142a0efba568fd69aa5def6c45b5dca07f 0.0s
=> => exporting config sha256:4530c43296a490b36a6bff8373a410fb3a9ed622713b98567ed982b29588df82 0.0s
=> => exporting attestation manifest sha256:734980ad27051ccaf0f0424da0f1fb90e5ebfe1df048320b331884fb6962bd 0.1s
=> => exporting manifest list sha256:a251b30682c6f83e3eb189f29ca60572d540f495058886dfc0a058be4907a6b5 0.0s
=> => naming to doc 0.0s
=> => unpacking to 1.1s
docker-desktop://dashboard/build/desktop-linux/desktop-linux/wxxhl77ypis6c0xhot1sxf9ns (ctrl + click)
View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/wxxhl77ypis6c0xhot1sxf9ns

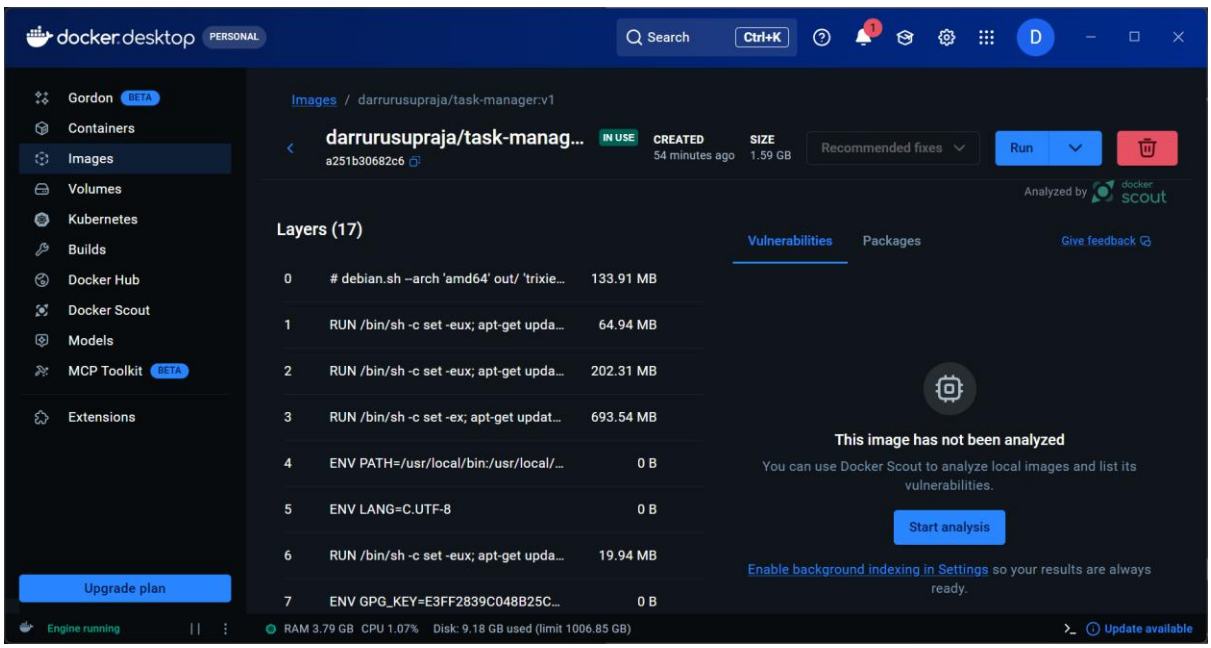

TERMINAL PORTS DEBUG CONSOLE
TERMINAL
docker
PS C:\Users\HP\OneDrive\Desktop\task-manager> docker run -p 5001:5000 darrurusupraja/task-manager:v1
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.17.0.2:5000
Press CTRL+C to quit
172.17.0.1 - - [11/Apr/2026 07:10:59] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [11/Apr/2026 07:10:59] "GET /favicon.ico HTTP/1.1" 404 -

```
PS C:\Users\HP\OneDrive\Desktop\task-manager> docker login
Authenticating with existing credentials... [Username: darrurusupraja]

i Info → To login with a different account, run 'docker logout' followed by 'docker login'


Login Succeeded
```

```
PS C:\Users\HP\OneDrive\Desktop\task-manager> docker push darrurusupraja/task-manager:v1
The push refers to repository [docker.io/darrurusupraja/task-manager]
6e0d85271a26: Already exists
79d5bd8a8d26: Pushed
89d573bf42b3: Layer already exists
081ccf923272: Layer already exists
26dfe2fac1c4: Pushed
795dbedde24d: Pushed
c9723aa529b0: Layer already exists
dd980b09fe9d: Pushed
599ba7a62378: Layer already exists
ee630ed9c559: Layer already exists
91c91c91f1d2: Layer already exists
v1: digest: sha256:a251b30682c6f83e3eb189f29ca60572d540f495058886dfc0a058be4907a6b5 size: 856
```




```
PS C:\Users\HP\task-manager> docker images
                                                                                    i Info →   U  In Use
IMAGE                                          ID            DISK USAGE   CONTENT SIZE   EXTRA
darrurusupraja/springboot-app:latest           21c5d4aded5b       674MB          216MB    U
darrurusupraja/task-manager:v1                 a251b30682c6       1.6GB          410MB    U
darrurusupraja/task-manager:v2                 260fd329ce98       1.6GB          411MB    U
darrurusupraja/task-manager:v3                 435233ec5a59       1.6GB          410MB    U
darruusupraja/task-manager:v1                  85af08f15209       1.6GB          410MB    U
docker/desktop-cloud-provider-kind:v0.5.0      4ad59ce20658       595MB          162MB    U
docker/desktop-containerd-registry-mirror:v0.0.3  179a9ea3ae47    50.3MB         16.1MB    U
docker/welcome-to-docker:latest                c4d56c24da4f      22.2MB         6.03MB    U
envoyproxy/envoy:v1.36.4                       08acc53db6f4       248MB         62.1MB
kindest/node:v1.34.3                           08497ee19eac      1.35GB          395MB    U
nginx:stable-alpine3.23-perl                   84885adffd68       150MB         38.1MB
springboot-app:latest                          6f635f9cc16f       674MB          216MB    U
supraja123/taskmanager:v1                      ec2775e678d1       1.6GB          411MB
task-manager:v1                                85af08f15209       1.6GB          410MB    U
taskmanager:v1                                 ec2775e678d1       1.6GB          411MB
yourdockerhub/taskmanager:v1                   31251c24a751       1.6GB          411MB
PS C:\Users\HP\task-manager>
```

Jenkins / All / New Item

New Item

Enter an item name

task-manager-pipeline

Select an item type

Pipeline
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.

Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

Multi-configuration project
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

Folder
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.

Multibranch Pipeline
Creates a set of Pipeline projects according to detected branches in one SCM repository.

OK

Configure

- General
- Triggers
- Pipeline
- Advanced

Definition

Pipeline script

Script ?

try sample Pipeline...

```
    }

    parameters {
        choice(name: 'VERSION', choices: ['v1.0', 'v2.0', 'v3.0'], description: 'Select version to deploy')
    }

    stages {

        stage('Checkout') {
            steps {
                // Checkout the main branch first so Jenkins handles the initialization properly
                git branch: 'main', url: 'https://github.com/DARRURUSUPRAJA/task-manager.git'
                // Force a checkout of the specific tag using native Git CLI!
                bat "git checkout tags/${params.VERSION}"
            }
```

Use Groovy Sandbox ?

Pipeline Syntax

Save Apply

```
PS C:\Users\HP\OneDrive\Desktop\task-manager> kubectl apply -f v1-deployment.yaml
deployment.apps/task-manager-v1 unchanged
service/task-manager-v1-service unchanged
PS C:\Users\HP\OneDrive\Desktop\task-manager> kubectl apply -f v2-deployment.yaml
deployment.apps/task-manager-v2 unchanged
service/task-manager-v2-service unchanged
PS C:\Users\HP\OneDrive\Desktop\task-manager> kubectl apply -f v3-deployment.yaml
deployment.apps/task-manager-v3 unchanged
service/task-manager-v3-service unchanged
PS C:\Users\HP\OneDrive\Desktop\task-manager> kubectl get pods
NAME                              READY   STATUS    RESTARTS   AGE
task-manager-v1-6d8dccd4c9-cs9nn  1/1     Running   0          16m
task-manager-v2-64fc59d8dc-k5zqv  1/1     Running   0          16m
task-manager-v3-67c8d548b7-kwdd2  1/1     Running   0          16m
PS C:\Users\HP\OneDrive\Desktop\task-manager> kubectl get services
NAME                     TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)          AGE
kubernetes               ClusterIP   10.96.0.1        <none>        443/TCP          31m
task-manager-v1-service  NodePort    10.110.111.148   <none>        5000:30001/TCP   17m
task-manager-v2-service  NodePort    10.106.103.155   <none>        5000:30002/TCP   17m
task-manager-v3-service  NodePort    10.96.85.155     <none>        5000:30003/TCP   17m
PS C:\Users\HP\OneDrive\Desktop\task-manager> minikube ip
192.168.49.2
```

```
PS C:\Users\HP\OneDrive\Desktop\task-manager> minikube service task-manager-service
```

NAMESPACE	NAME	TARGET PORT	URL
default	task-manager-service	5000	http://192.168.49.2:30007

```
🔗  Starting tunnel for service task-manager-service.
```

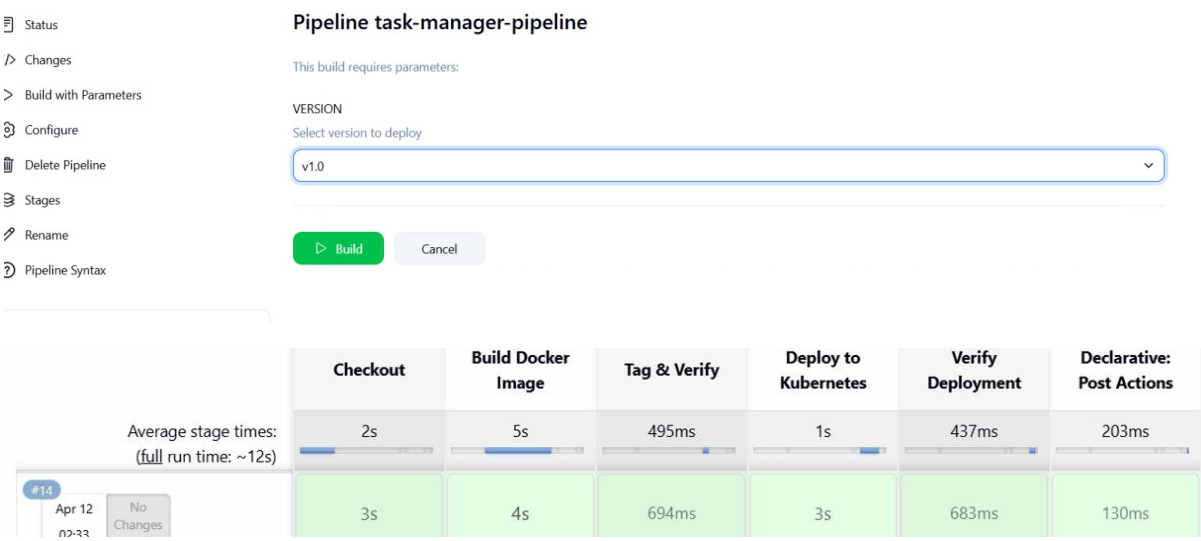
NAMESPACE	NAME	TARGET PORT	URL
default	task-manager-service		http://127.0.0.1:58943

```
🎉  Opening service default/task-manager-service in default browser...
❗  Because you are using a Docker driver on windows, the terminal needs to be open to run it.
```

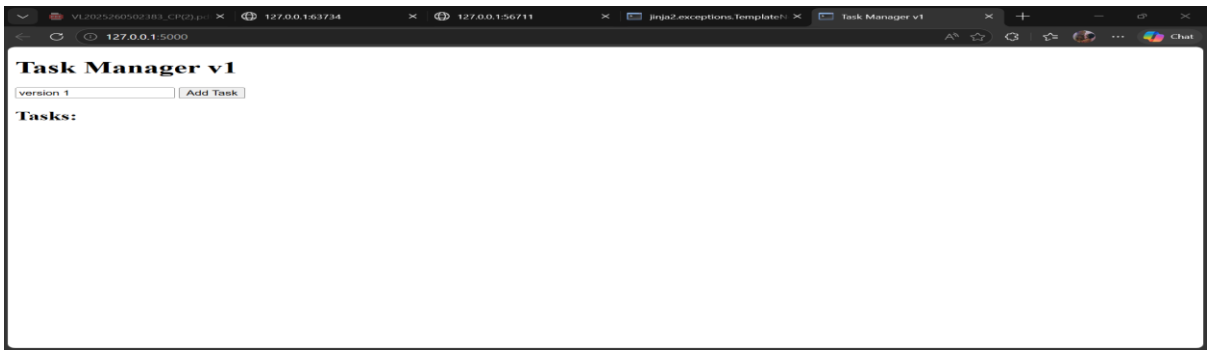
```
PS C:\Users\HP\OneDrive\Desktop\task-manager\v1> python app.py
 * Serving Flask app 'app'
 * Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI serv
er instead.
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
 * Restarting with stat
 * Debugger is active!
 * Debugger PIN: 244-693-218
```

task-manager	v1.0	934f5602ac61	7 hours ago	212.32 MB	
task-manager	v2.0	ac0cc7ccd0bf	7 hours ago	212.33 MB	
task-manager	v3.0	31044c34b597	7 hours ago	212.33 MB	

Version Swithching Using Parameter In Jenkins:



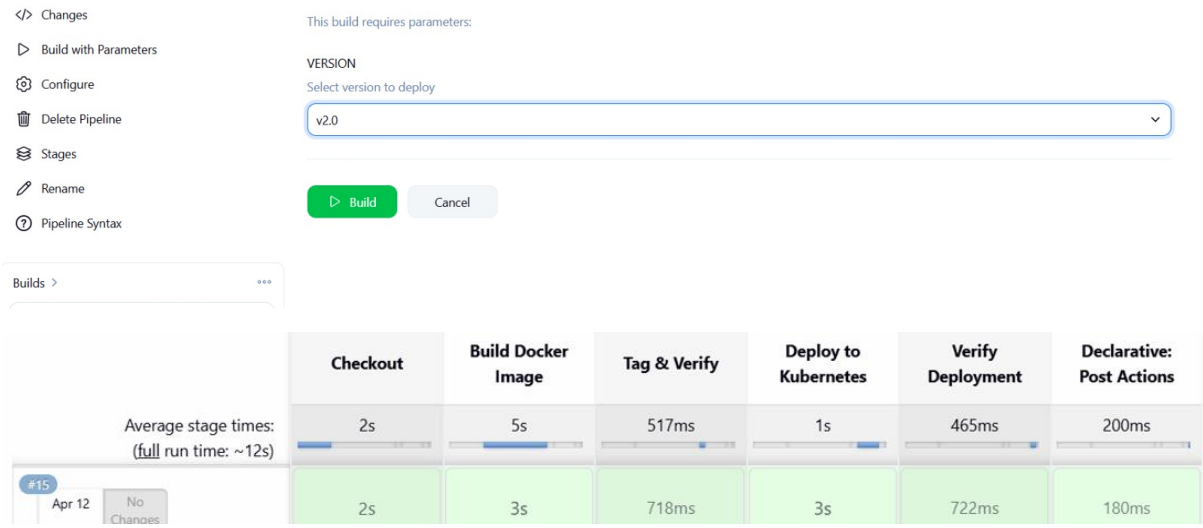
Output For Version 1:



Version-2

```
PS C:\Users\HP\OneDrive\Desktop\task-manager> docker build -t task-manager:v2 .
[+] Building 22.1s (10/10) FINISHED                                                    docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                   0.1s
 => => transferring dockerfile: 139B                                                                   0.1s
 => [internal] load metadata for docker.io/library/python:3.9                                          2.9s
 => [auth] library/python:pull token for registry-1.docker.io                                          0.0s
 => [internal] load .dockerignore                                                                      0.0s
 => => transferring context: 2B                                                                        0.0s
 => [1/4] FROM docker.io/library/python:3.9@sha256:da5aee29682d12a6649f51c8d6f15b87deb3e6c524b923c41d0cb3304d07c913   0.1s
 => => resolve docker.io/library/python:3.9@sha256:da5aee29682d12a6649f51c8d6f15b87deb3e6c524b923c41d0cb3304d07c913   0.1s
 => [internal] load build context                                                                      0.2s
 => => transferring context: 45.07kB                                                                   0.2s
 => CACHED [2/4] WORKDIR /app                                                                          0.0s
 => [3/4] COPY . .                                                                                     0.2s
 => [4/4] RUN pip install -r requirements.txt                                                         14.9s
 => exporting to image                                                                                 3.4s
 => => exporting layers                                                                                1.7s
 => => exporting manifest sha256:012862538b47fb16d697ab01895fb722e770b5e196e6f91f21d6ba6b235eac47      0.0s
 => => exporting config sha256:15f177ec0a6d4041db5a172f2f494d3f63e073e67798a4aa3255d0191f15f577        0.0s
 => => exporting attestation manifest sha256:8e8dacfa9056622902bf7e62fb84b13d41e7ce0368eca9bd46d64587b51a2cc0   0.1s
 => => exporting manifest list sha256:cce86d601938b1c3c6fb1a12ce4f863251cff59cf6dadc5013f7272c23ab7cc2   0.0s
 => => naming to docker.io/library/task-manager:v2                                                     0.0s
 => => unpacking to docker.io/library/task-manager:v2                                                  1.4s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/lvc7yq96ydysu632741uo4lnh
```




```
PS C:\Users\HP\OneDrive\Desktop\task-manager> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
task-manager-v1-578599d85b-rgs2h    1/1     Running   0          13m
task-manager-v2-64fc59d8dc-k5zqv    1/1     Running   0          54m
task-manager-v3-67c8d548b7-kwdd2    1/1     Running   0          54m
PS C:\Users\HP\OneDrive\Desktop\task-manager> minikube service task-manager-v2-service
```

NAMESPACE	NAME	TARGET PORT	URL
default	task-manager-v2-service	5000	http://192.168.49.2:30002

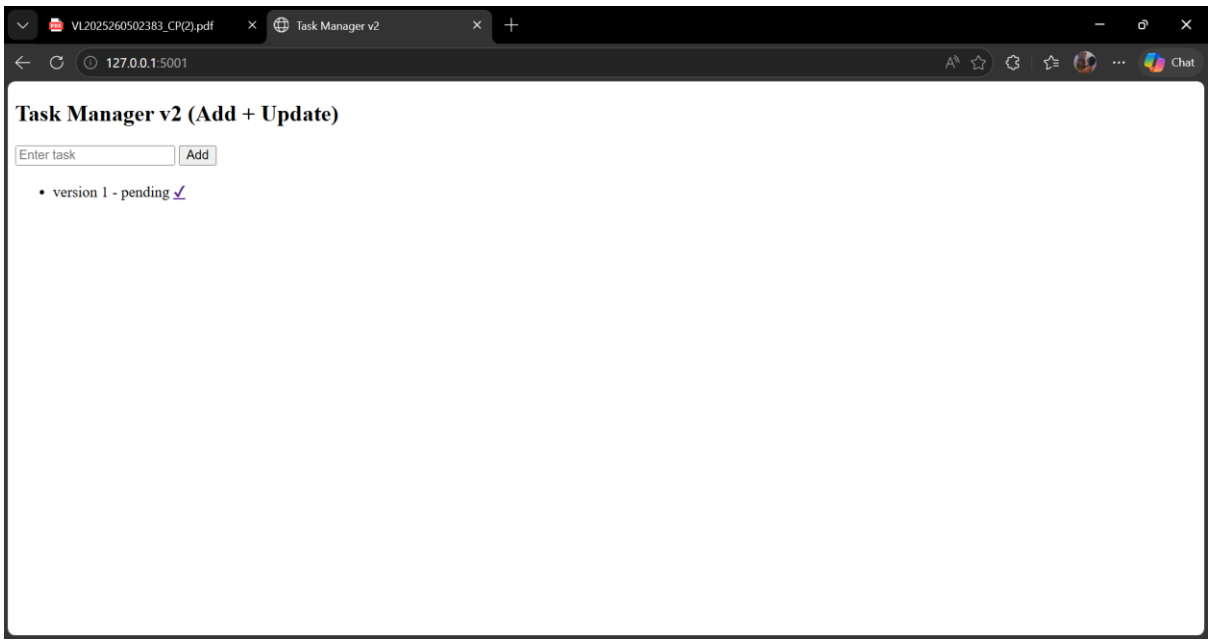
```
🔗  Starting tunnel for service task-manager-v2-service.
```

NAMESPACE	NAME	TARGET PORT	URL
default	task-manager-v2-service		http://127.0.0.1:60131

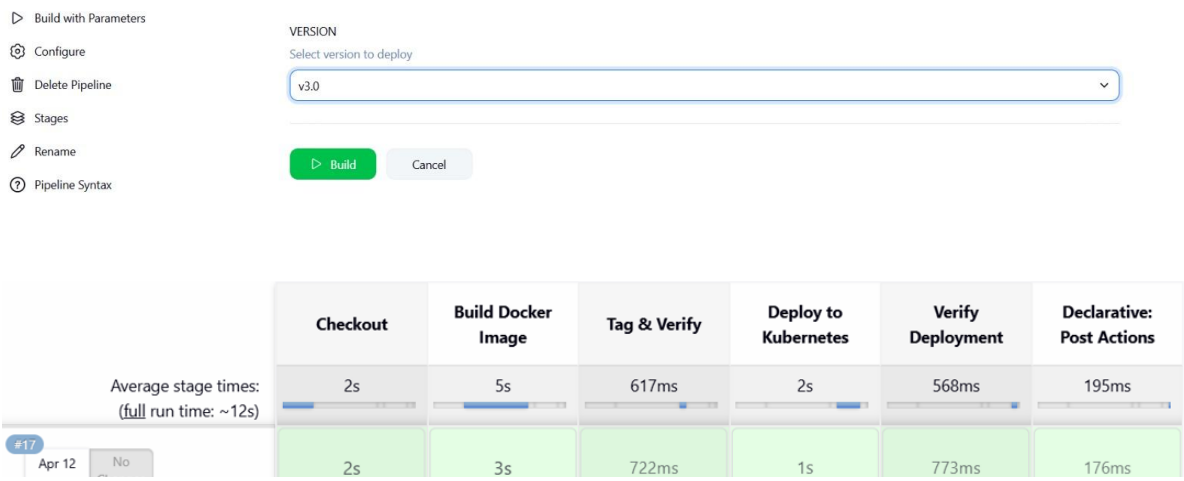
```
🎉  Opening service default/task-manager-v2-service in default browser...
❗  Because you are using a Docker driver on windows, the terminal needs to be open to run it.
```

```
PS C:\Users\HP\OneDrive\Desktop\task-manager\v2> python app.py
 * Serving Flask app 'app'
 * Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI serv
er instead.
 * Running on http://127.0.0.1:5001
Press CTRL+C to quit
 * Restarting with stat
 * Debugger is active!
 * Debugger PIN: 244-693-218
127.0.0.1 - - [11/Apr/2026 18:07:39] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [11/Apr/2026 18:07:51] "POST /add HTTP/1.1" 302 -
127.0.0.1 - - [11/Apr/2026 18:07:51] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [11/Apr/2026 18:08:36] "GET /update/0 HTTP/1.1" 302 -
127.0.0.1 - - [11/Apr/2026 18:08:36] "GET / HTTP/1.1" 200 -
```

Output For Version 2:


VL2025260502383_CP(2).pdf
Task Manager v2
127.0.0.1:5001
Chat
Task Manager v2 (Add + Update)
Enter task
Add
version 1 - pending

Task Manager v2
127.0.0.1:5001
Chat
Task Manager v2 (Add + Update)
Enter task
Add
version 1 - done


Build with Parameters
Configure
Delete Pipeline
Stages
Rename
Pipeline Syntax
VERSION
Select version to deploy
v3.0
Build
Cancel
Checkout
Build Docker Image
Tag & Verify
Deploy to Kubernetes
Verify Deployment
Declarative: Post Actions
Average stage times:
(full run time: ~12s)
2s
5s
617ms
2s
568ms
195ms
#17
Apr 12
No
2s
3s
722ms
1s
773ms
176ms

Version3:

```
PS C:\Users\HP\OneDrive\Desktop\task-manager> docker build -t task-manager:v3 .
[+] Building 2.6s (9/9) FINISHED                                                                    docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                0.0s
 => => transferring dockerfile: 139B                                                                                0.0s
 => [internal] load metadata for docker.io/library/python:3.9                                                       1.8s
 => [internal] load .dockerignore                                                                                   0.0s
 => => transferring context: 2B                                                                                     0.0s
 => [1/4] FROM docker.io/library/python:3.9@sha256:da5aee29682d12a6649f51c8d6f15b87deb3e6c524b923c41d0cb3304d07c913  0.1s
 => => resolve docker.io/library/python:3.9@sha256:da5aee29682d12a6649f51c8d6f15b87deb3e6c524b923c41d0cb3304d07c913  0.1s
 => [internal] load build context                                                                                   0.1s
 => => transferring context: 4.66kB                                                                                 0.0s
 => CACHED [2/4] WORKDIR /app                                                                                       0.0s
 => CACHED [3/4] COPY . .                                                                                           0.0s
 => CACHED [4/4] RUN pip install -r requirements.txt                                                                0.0s
 => exporting to image                                                                                              0.3s
 => => exporting layers                                                                                             0.0s
 => => exporting manifest sha256:012862538b47fb16d697ab01895fb722e770b5e196e6f91f21d6ba6b235eac47                   0.0s
 => => exporting config sha256:15f177ec0a6d4041db5a172f2f494d3f63e073e67798a4aa3255d0191f15f577                     0.0s
 => => exporting attestation manifest sha256:c59d2b3b1a6e6061eb5a8c71391cb6c568261cc1d5cd2a872686e6acd9eb04c1       0.1s
 => => exporting manifest list sha256:31da50a4ab34a828455ae63004d72db40ab08a49cc95142278c1b0853b95178f              0.0s
 => => naming to docker.io/library/task-manager:v3                                                                  0.0s
 => => unpacking to docker.io/library/task-manager:v3                                                               0.0s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/ldhgsalm419yiqr69eogj75oc
```

```
PS C:\Users\HP\OneDrive\Desktop\task-manager> minikube service task-manager-v3-service
```

NAMESPACE	NAME	TARGET PORT	URL
default	task-manager-v3-service	5000	http://192.168.49.2:30003

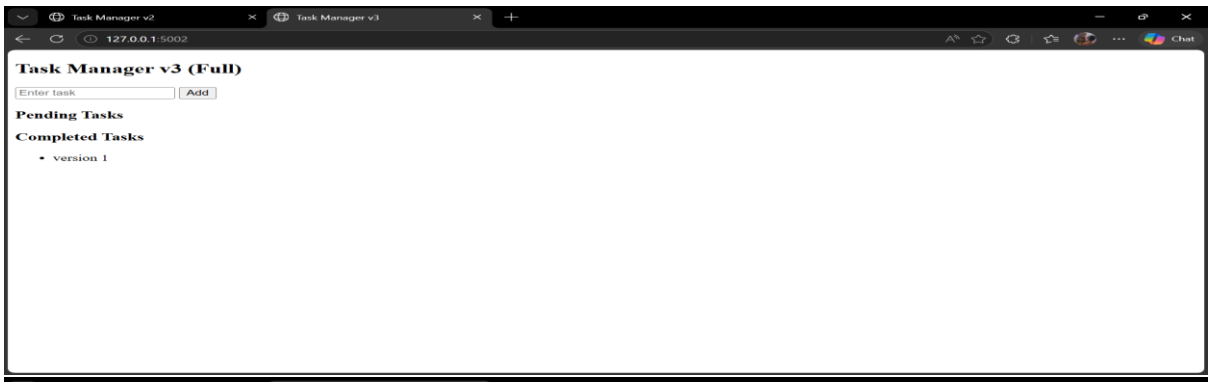
```
🔗  Starting tunnel for service task-manager-v3-service.
```

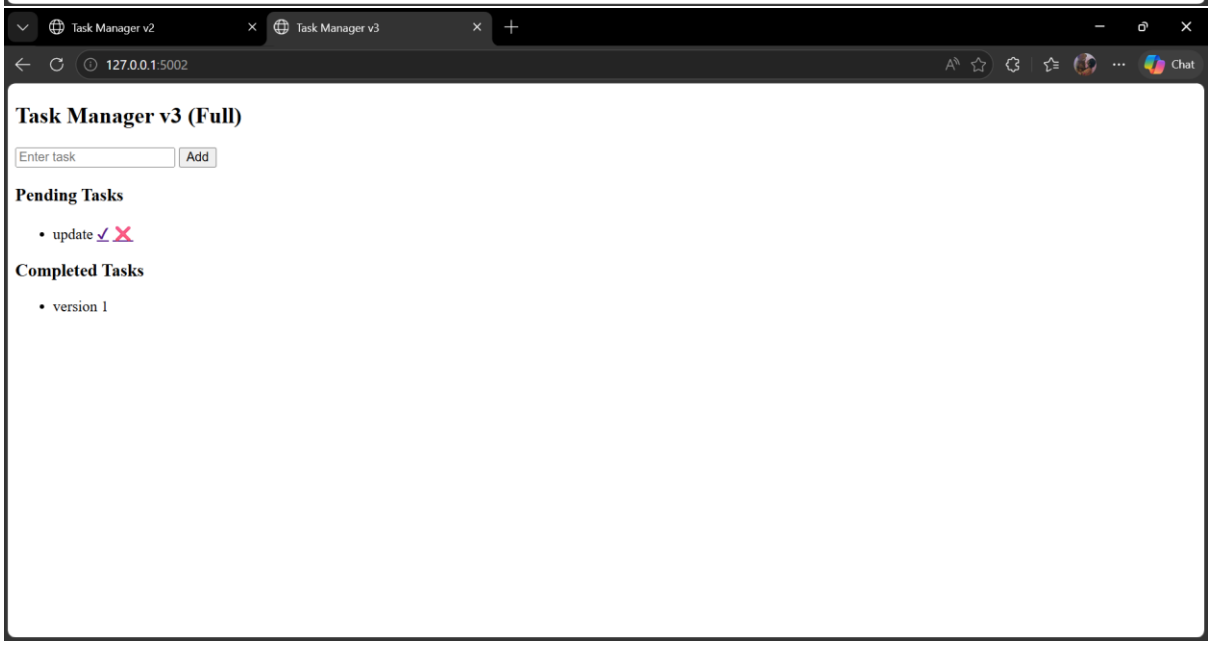
NAMESPACE	NAME	TARGET PORT	URL
default	task-manager-v3-service		http://127.0.0.1:54214

```
🎉  Opening service default/task-manager-v3-service in default browser...
❗  Because you are using a Docker driver on windows, the terminal needs to be open to run it.
```

```
PS C:\Users\HP\OneDrive\Desktop\task-manager\v3> python app.py
 * Serving Flask app 'app'
 * Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI serv
er instead.
 * Running on http://127.0.0.1:5002
Press CTRL+C to quit
 * Restarting with stat
 * Debugger is active!
 * Debugger PIN: 244-693-218
127.0.0.1 - - [11/Apr/2026 18:09:41] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [11/Apr/2026 18:09:46] "POST /add HTTP/1.1" 302 -
127.0.0.1 - - [11/Apr/2026 18:09:46] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [11/Apr/2026 18:10:17] "GET /complete/0 HTTP/1.1" 302 -
127.0.0.1 - - [11/Apr/2026 18:10:17] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [11/Apr/2026 18:11:15] "POST /add HTTP/1.1" 302 -
```

Output For Version 3:


Task Manager v2
Task Manager v3
127.0.0.1:5002
Chat
Task Manager v3 (Full)
Enter task
Add
Pending Tasks
Completed Tasks
version 1


Task Manager v2
Task Manager v3
127.0.0.1:5002
Chat
Task Manager v3 (Full)
Enter task
Add
Pending Tasks
update
Completed Tasks
version 1